

Project (Titanic: Machine Learning from Disaster)

Dataset: Titanic - Machine Learning from Disaster (Kaggle)

Attributes:

Attribute Name	Definition	Example
PassengerId	Unique ID assigned to each passenger	1, 2, 3
Survived (Target)	Indicates whether the passenger survived (0 = No, 1 = Yes)	0, 1
Pclass	Ticket class (1 = First, 2 = Second, 3 = Third)	1, 3
Name	Passenger's full name	Braund, Mr. Owen Harris
Sex	Gender of the passenger	Male, Female
Age	Age in years	22, 38
SibSp	Number of siblings/spouses aboard	0, 1, 3
Parch	Number of parents/children aboard	0, 1, 2
Ticket	Ticket number	A/5 21171
Fare	Ticket fare	7.25, 71.83
Cabin	Cabin number (may contain missing values)	C85
Embarked	Port of embarkation (C, Q, S)	C, S

Project Requirements

1-Preprocessing: Before building your classification models, you

need to make sure that the dataset is processed, clean and ready-

to-use.

- Handle missing values
- Encode categorical variables
- Feature scaling if needed

2-Classification

- Train at least 3 models (Logistic Regression, Decision Tree, SVM)
- Tune at least two hyperparameters
- Train the models on the given training set (train_data.csv) and evaluate the model on the given testing set (test_data.csv).

3-Report

You must explain in details the preprocessing techniques

that you applied on the dataset before the modeling step.

- Perform analysis and visualizations on the dataset and explain how features affect each other.
- State the models and hyperparameters used in the modeling step
- Mention any other techniques that you used to enhance the results.
- Write a final conclusion.

Note:

Additional techniques will be considered bonus.

A portion of the marks will be based on individual evaluation